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Abstract. Where should the question go in a vision-language model (VLM) prompt: before the image or after it? Intuition says before: knowing what is asked should tell the model where to look. Yet across visual question answering benchmarks, question-first prompting consistently underperforms the image-first ordering recommended for frontier VLMs, a phenomenon we term the *question-first paradox*. We trace the paradox to a conflict between two stages of VLM computation. Logit-lens and attention probes show the intuition is half right: a question placed before the image genuinely steers perception, moving image patch representations toward question-relevant concepts. The failure lies downstream. Stranded behind hundreds of image tokens, the question is barely attended by the answer token, which instead commits to image-driven (often wrong) answers; a causal attention knockout confirms that the answer reads the question only when the question follows the image. Better perception is squandered by worse access. The diagnosis yields a training-free fix: *question echoing*, restating the question on both sides of the image so that one copy steers perception while the other is read out at answer time. A similar division of labor appears in a fifty-year-old finding on human “adjunct questions”, where repeating a question before and after a passage aids comprehension more than either position alone. *Echoing the image* as well brings further gains, restoring the whole-image view a causal decoder otherwise loses. The paradox holds across five open VLMs, costing up to 17.5 group-accuracy points. Echoed prompts recover most of the gap and, on the harder benchmarks (NaturalBench and Winoground), surpass the best single-pass ordering by up to 19 group-accuracy points on Winoground, with no training, fine-tuning, or architecture change. The paradox exposes a tension in multimodal prompting between steering what a model *sees* and preserving access to what it was *asked*; echoing resolves it with prompt design alone.

Keywords: Vision-Language Models · Prompt Ordering · Visual Steering · Logit Lens · Question-First Paradox · Interpretability

1 Introduction

Every VLM prompt makes a silent decision: the question either precedes or follows the image. Modern vision-language models consume a single interleaved sequence of system, image, and task tokens, so the ordering is a free design choice, and today it is decided by folklore. Most model cards recommend image-first [1,2,7,17,19–21], while other prompting practices vary. A natural hypothesis,

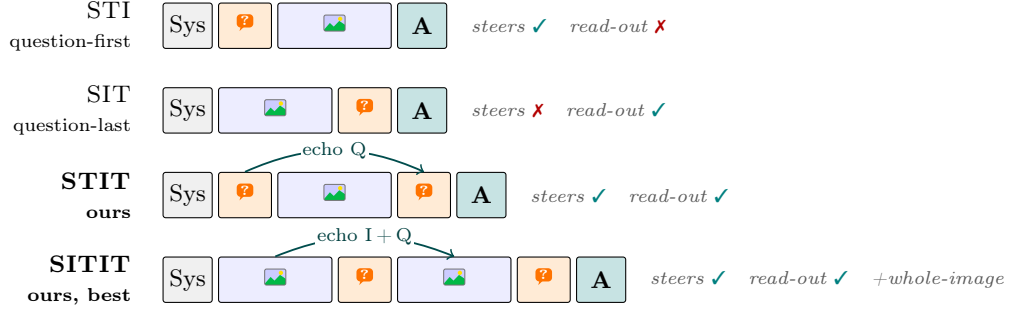


Fig. 1: Question placement, not question presence, decides the answer. A VLM reads system (Sys), image (🖼️), and question (❓) tokens as one stream and generates from the final answer position (A). Each row is marked on the two mechanisms a prompt must serve: whether the question *steers* what the image encodes, and whether it is *read out* at the answer. Question-first (STI) steers but is not read out: it strands the question before a long image span, so the answer barely attends to it and commits early to an image-anchored, often-wrong response. Question-last (SIT) is read out but does not steer, recovering most of the loss. Training-free *question echoing* (STIT) re-states the question on both sides of the image, keeping the steering while restoring question-answer adjacency; re-presenting the image with it (SITIT) also grants the causal decoder a whole-image view and gives the best accuracy.

borrowed from how people inspect a scene [6], says the folklore has it backwards: the question should come first, because a model that knows what is asked can decide where to look and what to encode.

The hypothesis fails, and it fails in an instructive way. Consider the two orderings that hold content fixed and move only the question: STI (System, Task, Image) places the question before the image, SIT (System, Image, Task) after. On NaturalBench with Qwen3-VL-8B, question-first scores 8.1 group-accuracy points below question-last (0.270 vs. 0.351). The same gap appears on POPE and Winoground and on three further VLM families (Qwen2.5-VL, InternVL3, LLaVA), reaching 17.5 points; it is muted only on Gemma-3-27B. We call this the *question-first paradox*: the ordering that should inform perception is the one that hurts (Fig. 1).

This paper opens the model to explain the paradox, then turns the explanation into a fix. The resolution is a dissociation between two stages of computation that prompt design conflates. Using the logit lens [3] and layer-wise probes, we show that question-first prompting *does* steer perception: image patches decode to more question-relevant concepts, and the answer position attends more strongly to the image. The steering is real, and it is wasted. Stranded before hundreds of image tokens, the question is under-read by the decoder, and the answer position commits early to an image-anchored, often wrong token before a question representation reaches it. **Better perception is squandered by worse access.**

Once stated this way, the fix writes itself. *Question echoing* (STIT) restates the question once before and once after the image: the first copy steers, the sec-

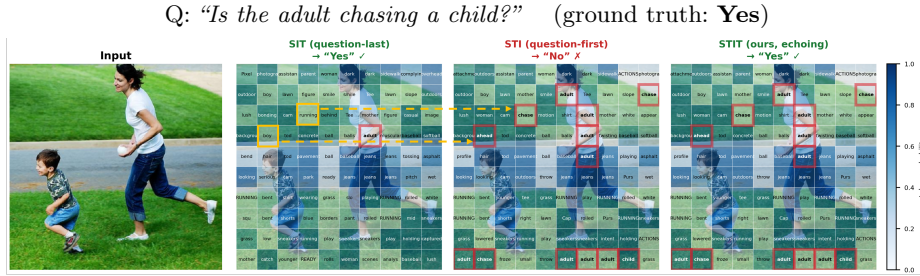


Fig. 2: Steering happens, yet is not read out. Each cell shows the vocabulary token its image patch’s final-layer (layer 32) hidden state decodes to (logit lens, Qwen3-VL), coloured by its probability (blue scale); red **rings** mark patches whose token reflects the question (*adult*, *chase*, *child*, *ahead*). Question-last (SIT) leaves the patches generic and answers correctly; question-first (STI) steers the *same* patches toward the question, turning *boy* and *running* into *ahead* and *chase*, yet answers wrong: steering happens, but is not read out. Our fix (echoing, STIT; Sec. 5) keeps the steering and answers correctly.

ond restores question–answer adjacency. *Image echoing* (SITIT) re-presents the image alongside the repeated question; under a causal mask every patch of the second copy attends over the entire first copy, recovering the bidirectional whole-image read that the vision encoder computes but the causal decoder discards. Both are pure prompt edits: no fine-tuning, no decoding change, no architecture change.

Strikingly, the fix has a fifty-year-old precedent in human learners. In educational psychology’s *adjunct question* literature, repeating the same question before and after a prose passage aids comprehension more than either position alone: the prequestion directs attention, the postquestion consolidates at test time [5, 9]. This is the division of labor our probes uncover inside the model, suggesting the two-mechanism structure of echoing is a general property of sequential comprehension rather than an artifact of one architecture.

Contributions. We make four contributions. First, we establish the *question-first paradox*: with content held fixed, placing the question before the image degrades accuracy across three benchmarks and five open VLMs, against the intuition that it should help (Sec. 3). Second, we diagnose it, showing that question-first prompting genuinely steers perception yet is not read out, and confirming with a causal attention knockout that the answer is computed from the question only when the question follows the image; perceptual steering and decoder access are dissociable, and the paradox lives entirely in access (Sec. 4). Third, we turn the diagnosis into a training-free fix, *question echoing*, which restates the question after the image to restore read-out while preserving the steering, together with an *image-echoing* variant that recovers a whole-image read under the causal mask (Secs. 5 and 5.2). Finally, we ablate the design, ruling out “more image tokens” and “more pre-answer compute” and isolating question adjacency and in-distribution image-first pairing as the active ingredients (Sec. 7).

The practical message: one duplicated question line recovers most of the question-first gap at no real cost, and a duplicated (image, question) pair exceeds the best single-pass ordering. The scientific message: visual steering and answer read-out are distinct mechanisms with different positional preferences, and prompt design must serve both. Prompt ordering can be a measured, mechanistically grounded science rather than folklore; this paper is a case study in making it one. Our code and a project website are included in the supplementary material.

2 Related Work

Prompt ordering in VLMs. Instruction-tuned VLMs such as LLaVA [15], Qwen2.5-VL [4], and Gemma 3 [24] flatten image and text into a single token stream consumed by a causal decoder, so the relative placement of visual tokens and the question is an unavoidable design choice. It is rarely treated as a variable in its own right: order-sensitivity studies focus on in-context demonstrations [23] or report image-first versus text-first accuracy as an unexplained ablation [11], and model cards recommend image-first [7, 17, 19–21] without saying why; no prior work establishes *why* question-first hurts. Concurrent work by Gupta and Gandelsman [8] independently documents the same failure and localizes it with activation patching to a narrow mid-network region, then closes it with asymmetric test-time training. We treat prompt ordering as a first-class object and differ in two ways: we localize a mechanism rather than a region, causally dissociating intact perceptual steering (Sec. 4.1) from a failed read-out (Sec. 4.3), and our fix is training-free (question echoing, Sec. 5).

Question repetition and re-reading. In text-only LLMs, re-presenting the query has a mixed record: re-reading prompts improve reasoning [26], repeated question tokens draw more attention [10], and prompt repetition helps most when the query sits far from the answer position [12], yet controlled studies also find the gains insignificant [22]. These works neither involve vision nor explain when repetition should help. In VLMs the long image span makes the query–answer distance structural rather than incidental; we identify decoder under-reading of the distant question as the mechanism and show echoing yields large, consistent gains exactly where the text-only literature is equivocal. Image echoing has no text-only analogue. The failure we document is related to “lost in the middle” [16], where decoder-only models under-use context far from the query, but with a twist the text-only setting lacks: the distant question still steers perception successfully, and the failure arises only at answer generation.

Interpreting VLM representations. The logit lens [3] projects intermediate hidden states through the output embedding to read what each layer represents; applied to VLMs, it shows that visual tokens progressively align with interpretable vocabulary [18]. We build on this tool but ask a different question: not what visual tokens encode, but how prompt ordering changes what they encode and whether the decoder reads it out.

3 The Question-First Paradox

3.1 Notation and setup

A prompt has three sections: the system message (S), the image (I, expanded into many visual tokens), and the task or question (T). We write an ordering as the sequence of section letters in token order; generation begins after the last letter. The system message is a fixed prefix prepended by default (Sec. H justifies this choice). The two single-question orderings hold content fixed and move only the question:

$$\text{STI} = \text{System, Task, Image} \quad (\text{question-first}), \quad (1)$$

$$\text{SIT} = \text{System, Image, Task} \quad (\text{question-last}). \quad (2)$$

They are anagrams. STI separates the question from the answer position by the entire image span; SIT places it immediately before the answer.

All orderings are produced by one order-aware input builder shared across models, so variants differ only in the arrangement of identical content. We evaluate on NaturalBench [13], POPE [14], and Winoground [25] (Sec. 2), with greedy decoding, a 16-token answer budget, and a fixed system prompt (Sec. A). Qwen3-VL-8B (primary, with all ablations) and Gemma-3-27B carry the full analysis; three further families (Qwen2.5-VL, InternVL3, LLaVA-1.5) test generality. We headline the most demanding metric, group accuracy on NaturalBench and Winoground, which a single wrong pair drives to zero and on which language priors do not help. For adjacent orderings we report paired significance over per-group correctness (two-sided exact McNemar; paired bootstrap 95% CI, 5,000 resamples).

3.2 The phenomenon

Table 1 isolates the paradox: hold the content fixed, move only the question, and question-first loses. On our primary model the gap is 8.1 group-accuracy points on NaturalBench and larger still on the compositional Winoground, all highly significant and free, since the token count is identical (per-comparison tests in Sec. B). The effect is not a quirk of one model. It reproduces on four of five VLMs and on all three benchmarks, and it is largest on the weaker models, where question-first can collapse a model to a single constant answer (Tab. 1). The lone exception is Gemma-3-27B, where the two orderings are statistically indistinguishable; even there the remedy of Sec. 6 still helps, so what generalizes is not the deficit but its cure.

If seeing the question first informs visual encoding, why does it hurt? Two families of explanation come to mind, and both turn out to be wrong: perhaps the steering simply does not happen (the image tokens are encoded the same regardless of what precedes them), or perhaps it happens and is harmful (question-conditioned encoding discards evidence the answer needs). Section 4 rejects both. The steering happens, it is benign, and it is thrown away.

Table 1: The question-first paradox, across architectures. Identical content; only the question moves. NaturalBench group accuracy for question-first (STI) vs. question-last (SIT), and the paradox gap $\Delta = \text{SIT} - \text{STI}$ on each benchmark (positive Δ means question-first is worse); **bold** marks each model’s largest gap. Paired significance for every gap is in Tab. 7. The paradox is large and consistent on four models and muted only on Gemma. All three benchmarks show the same paradox on the added models. It is most dramatic on LLaVA-1.5, where question-first collapses to a constant “Yes” on every benchmark (NaturalBench group 0.000, POPE accuracy 0.500, Winoground group 0.000) and question-last repairs it (POPE 0.871), the largest POPE gap of any model (+0.371). The Qwen2.5-VL and InternVL3 gaps (POPE up to +0.057, Winoground up to +0.210) also exceed the two primary models.

Model	NatBench acc paradox gap $\Delta = \text{SIT} - \text{STI}$				
	STI	SIT	NatBench	POPE	Wino
Qwen3-VL-8B	0.270	0.351	+0.081	+0.021	+0.095
Qwen2.5-VL-7B	0.102	0.277	+0.175	+0.057	+0.185
InternVL3-8B	0.276	0.355	+0.079	+0.053	+0.210
LLaVA-1.5-7B	0.000	0.123	+0.123	+0.371	+0.048
Gemma-3-27B	0.232	0.226	−0.006	+0.005	−0.005

Finding. Question *position*, not presence, drives a large accuracy gap: with identical content, question-first trails question-last by 8.1 NaturalBench and 9.5 Winoground group-accuracy points on Qwen3-VL-8B.

4 Anatomy of the Paradox

We open the model with two probes on the orderings that define the paradox, question-first (STI) and question-last (SIT). The *perception probe* asks what the image tokens encode, decoding each patch’s hidden state with the logit lens and measuring how far it moves from an image-only baseline. The *read-out probe* asks what the answer position uses, tracking layer by layer its attention over the question and image spans and the emergence of the correct answer token. The paradox resolves into a clean dissociation: steering lives in the first probe, the failure in the second. Unless noted, the correlational per-layer probes run on NaturalBench *disagreement pairs*: 150 yes/no questions that question-last answers correctly and question-first answers wrong (Sec. A).

4.1 Perception probe: steering is real and benign

Under STI, image patches decode to more question-relevant concepts than under a question-last ordering (Fig. 2). In the example the question asks whether an adult is chasing a child; question-first turns generic patch decodings into question-aligned ones (*adult*, *chase*, *child*). The attention probe agrees (Fig. 4, middle): under STI the answer position places *more* mass on the image span than question-last does (peak 0.237 vs. 0.094 for SIT). By the naive hypothesis

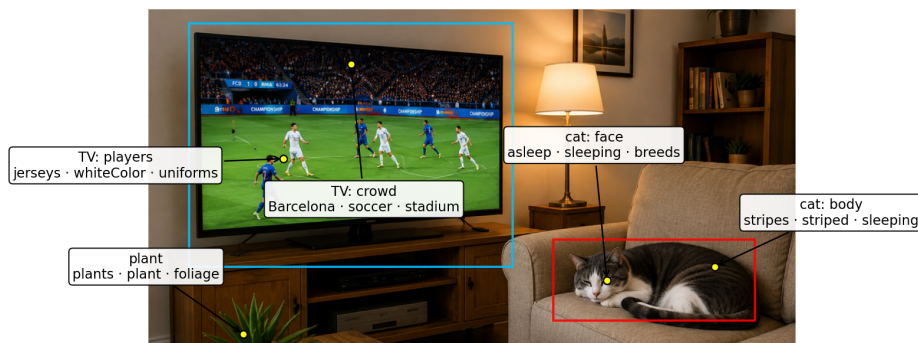


Fig. 3: The visual representation carries localized, readable object identity. Per-patch logit-lens decodings (Qwen3-VL, layer 28): each region decodes to its own object (crowd to *soccer*, cat to *asleep/stripped*, plant to *foliage*). Perception is correct and localized, so the question-first paradox is a downstream read-out failure (full analysis in Fig. 12).

this should help. It does not: STI is the worst ordering in every table in this paper, and the steered model still answers wrong.

We can measure the steering directly (Fig. 7). Relative to the image shown alone, question-last (SIT) barely moves the patches (mean cosine 0.91); the small residual is the system prompt that precedes the image, not the question, which SIT places after it and which the causal mask therefore hides from the patches. Question-first (STI) moves them markedly (mean 0.86, down to 0.62 in the most affected patches), a spatially structured shift. A question before the image reshapes perception; a question after it cannot.

This steered perception is also correct and well localized: decoded through the logit lens at a late layer, every region reports its own object (Fig. 3). The representation the read-out stage must consult is right and localized, so the paradox is a failure to consult it, not to form it.

4.2 Read-out probe: the question is under-attended and the decision locks in early

The read-out probe resolves the paradox (Fig. 4, left). The answer’s attention to the question peaks at only 0.068 under STI against 0.148 under SIT: separated from the answer by the whole image span, the question is barely read. The emergence curve (Fig. 4, right) tells the downstream story: the probability of the correct answer token rises with depth under question-last (final 0.85) but stays flat and low under question-first (final 0.14).

The decision-layer view sharpens this into timing. Tracking the two-way $P(\text{correct})$ over depth, STI commits to a wrong, image-anchored token in 41% of these pairs against 2% under SIT: question-first decides early, from the image, before a question representation reaches the answer.

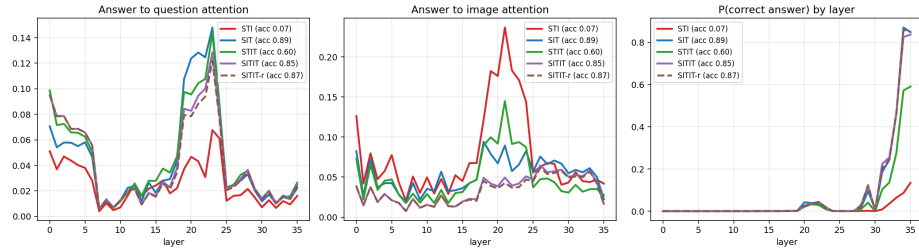


Fig. 4: Per-layer probe on NaturalBench disagreement pairs (STI-wrong, question-last-right; 150 pairs), averaged over pairs. *Left:* answer-to-question attention; STI (red) barely attends to the far-away question, while SIT, STIT, and SITIT do. *Middle:* answer-to-image attention; STI over-attends to the image. *Right:* logit-lens $P(\text{correct answer token})$ by layer; the correct answer emerges only for the orderings that place a question next to the answer. The re-representation variants of Sec. 5 (SITIT, dashed SITIT-R) track echoing on these probes.

Finding. Question-first fails at read-out, not perception. The answer under-attends the far-away question (peak 0.068 vs. 0.148 for question-last) and locks onto a wrong image-anchored token in 41% of disagreement pairs (vs. 2% for question-last); question position, not presence, decides what the answer is read from.

4.3 Causal intervention: the read-out edge is necessary

The probes above are correlational: they show the answer *attends less* to the question under STI, not that this edge *causes* the gap. We test causation with an attention knockout. During a single forward pass we add a pre-softmax $-\infty$ bias to every attention edge from the answer position to a chosen token span, at all 36 text layers, severing the answer’s ability to read that span while leaving every other computation intact. On 250 outcome-independent yes/no pairs we knock out, in turn, the *question* span, the *image* span, and a same-size *random* text span (a specificity control), under each ordering (Tab. 2). The knockout leaves the question in the prompt and cuts only the answer position’s *direct* edge to it: the question is still encoded and still read by every other position, and its content can still reach the answer indirectly through the tokens between the question and the answer, which have already attended to it. The knockout thus isolates the answer’s direct read-out of the question, not its total access, which is why severing an edge the answer does use lowers accuracy only partially rather than to chance (Sec. D).

The result is a clean double dissociation. Under question-last (SIT), cutting the answer’s edge to the question lowers accuracy by 0.056 (95% CI [0.020, 0.096], paired bootstrap $p \approx 0.003$), while cutting its edge to the image does nothing (0.000, n.s.): the answer is causally reading the question. Under question-first (STI) the dependence flips: the question edge is inert (+0.004, n.s.) and the *image* edge is what matters (0.096, [0.040, 0.152], $p \approx 0.001$). The random-span control is null in both orderings (≤ 0.008), and both interaction terms are significant (question: 0.060, $p \approx 0.006$; image: 0.096, $p \approx 0.001$). Widening the

Table 2: Causal attention knockout (Qwen3-VL-8B, 250 yes/no pairs). Change in per-pair accuracy when the answer’s attention to a span is severed at all layers (clean acc: STI 0.576, SIT 0.588); negative means the edge was used. Under question-last the answer causally reads the *question*; under question-first, the *image*. *: paired-bootstrap 95% CI excludes zero.

Severed edge	STI (Q-first)	SIT (Q-last)
answer $\rightarrow$ question	+0.004	− 0.056 *
answer $\rightarrow$ image	− 0.096 *	+0.000
answer $\rightarrow$ random	+0.004	−0.008

severed region to the whole post-question span reproduces the image dependence more starkly (STI image knockout -0.204). Question position does not merely change where the model *looks*; it changes what the answer is *computed from*.

Finding. The read-out edge is causal, not just correlational. Severing the answer’s attention to the question costs 0.056 accuracy under question-last ($p \approx 0.003$) but nothing under question-first, where the image edge carries the answer instead (0.096, $p \approx 0.001$); same-size random knockouts are null. Question position controls what the answer is computed from.

4.4 The mechanism predicts: the gap scales with question-answer distance

The read-out account makes a quantitative prediction. Under STI the image span separates the question from the answer, so the more vision tokens the image occupies, the further the question sits and the less it is read; under SIT the question is adjacent regardless of image size. The STI-SIT gap should therefore widen with image tokens. We test this on Qwen3-VL by re-rendering Natural-Bench at a range of resolutions, which sets the vision-token count, using only images large enough that every point is a genuine downscale. The prediction holds: the gap widens from 0.05 at 64 tokens to 0.08 at 324, then plateaus as question-last saturates near 0.35 once the image is legible while question-first keeps climbing (Sec. F). The diagnosis predicts the direction and the saturation, not merely the sign.

Together, the two probes replace the paradox with a mechanism. Question-first prompting buys perception steering at the price of decoder access, and in current VLMs the access matters more. The ideal prompt should not choose between the two: it should supply the question twice.

5 From Diagnosis to Prompt: Echoing

The diagnosis prescribes the prompt. Steering requires a question *before* the image (Sec. 4.1); read-out requires a question *adjacent to the answer* (Sec. 4.2). No single question copy can be in both places, so we use two.

5.1 Question echoing (STIT)

Our base prompt restates the question on both sides of the image:

$$\text{STIT} = \text{System, Task, Image, Task.} \quad (3)$$

The pre-image copy provides the top-down signal that steers visual encoding (the promise of question-first). The post-image copy restores question-answer adjacency (the strength of question-last). Echoing adds only the token cost of one repeated question, a dozen tokens against hundreds of image tokens.

The Sec. 4 probes confirm echoing realizes both halves of the diagnosis. Its pre-image copy reproduces question-first’s perceptual steering exactly: the image sees the same tokens before it, so under the causal mask its patch encodings coincide with STI’s (Fig. 7, cosine 1.000), and the two decode identically (Fig. 2). Its post-image copy restores read-out: the answer now attends to the adjacent question (Fig. 4; peak 0.144, against STI’s 0.068), the correct answer re-emerges with depth (final $P(\text{correct})$ 0.59, against 0.14), and the model commits early to a wrong token in only 17% of the disagreement pairs, against STI’s 41%. Echoing keeps the steering of question-first and the read-out of question-last.

5.2 Echoing the image (SITIT, SITIT-R)

A stronger variant *echoes the image* as well, re-presenting it alongside the repeated question:

$$\text{SITIT} = \text{System, Image, Task, Image, Task.} \quad (4)$$

The model sees the (image, question) pair twice and answers adjacent to the second question. Two effects combine. First, the second pass re-encodes the image with the question already in context, yielding question-specific features. Second, it removes a causal-attention bottleneck. The vision encoder is bidirectional over patches, but the Qwen3-VL decoder attends to those patch tokens causally, so within the decoder a patch cannot draw on patches that follow it (Fig. 5, left). Repetition alone removes the limitation: under the causal mask, every patch of the second copy attends over the *entire* first copy (Fig. 5, middle), so its representation integrates the whole image, a bidirectional-style read obtained without changing the architecture.

Read as a whole, SITIT combines the strengths of both baselines: its opening image-question pair reproduces SIT, the trained format, encoded in-distribution, and its second image is the question-steered pass of STI (Fig. 7), now attending over the whole first copy and sitting adjacent to the answer, where the decoder can read it instead of stranding it.

A further variant reverses the second copy’s patch order and positional indices (SITIT-R); Sec. C shows its effect is small and follows the decoder’s attention mask.

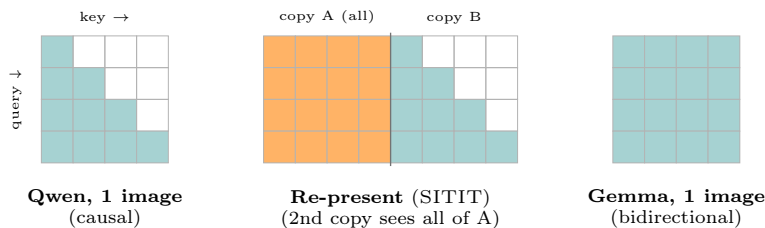


Fig. 5: The attention reachability the mechanism turns on. A filled cell means the query patch (row) can attend to the key patch (column). *Left:* in Qwen’s causal decoder a single image lets patch k attend only to $1 \dots k$, so its representation misses later patches. *Middle:* re-presenting the image (SITIT) makes every copy-B patch attend over all of copy A (orange), so it integrates the whole image, a bidirectional read produced by repetition alone. *Right:* Gemma’s decoder already attends bidirectionally within an image block, so repetition adds less. Reversing copy B (SITIT-R) keeps the orange whole-image read and only flips which copy-B neighbours (teal) refine each patch, which is why it changes little.

6 Results

6.1 The position ladder

Figure 6 plots the NaturalBench position ladder across all five models, and Tab. 3 adds POPE and Winoground. The results form a ladder. On Qwen3-VL, moving the question after the image recovers 8.1 NaturalBench group-accuracy points with the same tokens; echoing matches that question-last baseline, and re-presenting the image is best, a significant +0.024 over echoing and 10.4 points over question-first, with every sub-metric in the same order (Tab. 8). The pattern holds across models, with two honest exceptions the figure makes visible. On Gemma-3, where question order does not matter, echoing still helps significantly on NaturalBench and Winoground (near-saturated POPE stays flat), so the fix is not merely undoing a Qwen-family quirk. On Qwen2.5-VL, question echoing alone under-recovers, but image re-presentation restores it. LLaVA-1.5 is the extreme case: question-first collapses to a constant “Yes” on every benchmark and question-last repairs it, its POPE gap (+0.371) the largest of any model; being single-image, it has no image-re-presentation ordering.

The ladder is steepest on compositional Winoground: Qwen3-VL climbs from 0.223 to 0.410 group accuracy along the ladder, a 19-point gain over question-first ($p < 10^{-13}$), and Gemma-3 from 0.213 to 0.318. On near-saturated POPE every ordering except question-first sits within a point of the best, so we do not over-read it.

The two echoed variants carry different costs and different wins. Question echoing (STIT), which adds no image tokens, recovers the gap and beats question-last on NaturalBench and Winoground, but only matches it on near-saturated POPE and slightly trails it on VQAv2 (Sec. 6.1); its clean win is the compositional benchmarks. Image re-presentation (SITIT) is uniformly best but adds a second image, roughly doubling image prefill (Sec. 6.2). One model resists the

Table 3: Main results (primary models). NaturalBench group accuracy (1,900 groups), POPE accuracy/F1 (9,000 questions), and Winoground text/group accuracy (400 groups) for the two primary models. Question echoing (STIT) recovers the question-first gap and image re-presentation (SITIT, SITIT-R) matches or exceeds question-last, winning most on the hardest benchmark (Winoground). On Qwen3-VL the STI→SIT gap and the STIT→SITIT gain are significant (McNemar $p < 10^{-12}$, $p = 0.016$); on Gemma STI≈SIT ($p = 0.62$) yet echoing still helps. Best per model in **bold**. Figure 6 shows the NaturalBench ladder for all five models.

Model	Ordering	NatBench	POPE		Winoground	
		Group	Acc	F1	Text	Group
Qwen3-VL-8B	STI (Q-first)	0.270	0.870	0.861	0.699	0.223
	SIT (Q-last)	0.351	0.891	0.884	0.743	0.318
	STIT (ours)	0.350	0.884	0.876	0.774	0.375
	SITIT (ours)	0.374	0.889	0.881	0.779	0.403
	SITIT-R (ours)	0.374	0.889	0.883	0.781	0.410
Gemma-3-27B	STI (Q-first)	0.232	0.840	0.830	0.680	0.213
	SIT (Q-last)	0.226	0.845	0.845	0.683	0.208
	STIT (ours)	0.253	0.838	0.832	0.708	0.253
	SITIT (ours)	0.255	0.834	0.836	0.733	0.318
	SITIT-R (ours)	0.252	0.835	0.837	0.721	0.298

free fix: on Qwen2.5-VL, STIT alone under-recovers and only the image copy reaches question-last (Tab. 8). The free variant is a strong default, not a universal one.

Finding. The gains are largest where the task is hardest. On Winoground group accuracy our prompts beat the best baseline by 9.2 points on Qwen3-VL (0.410 vs. 0.318, $p < 10^{-5}$) and 11.0 on Gemma-3 (0.318 vs. 0.208, $p < 10^{-6}$); on near-saturated POPE all orderings except question-first are within a point.

Open-ended generation. Every benchmark so far is yes/no or forced-choice, so one might worry the ladder is an artifact of a constrained output space. It is not: on 2,000 VQAv2 validation questions scored with the official soft-accuracy metric over free-form generation, the ladder survives (question-first 0.811, question-last 0.834, image re-presentation best at 0.838; the STI-SIT gap excludes zero at 95%; Tab. 4). On this open-ended metric question echoing alone recovers part of the gap (0.821) but does not surpass question-last; the win over question-last comes from re-presenting the image. The paradox itself is a property of where the question sits, not of the answer format.

6.2 The cost of each ordering

The two echoed variants carry very different inference cost, a deployment fact worth stating. A VLM prefills the whole prompt before decoding, so cost tracks the token count: the KV-cache grows linearly and self-attention prefill quadratically. Image tokens dominate (Tab. 5: 2,305 per image against about 65 for

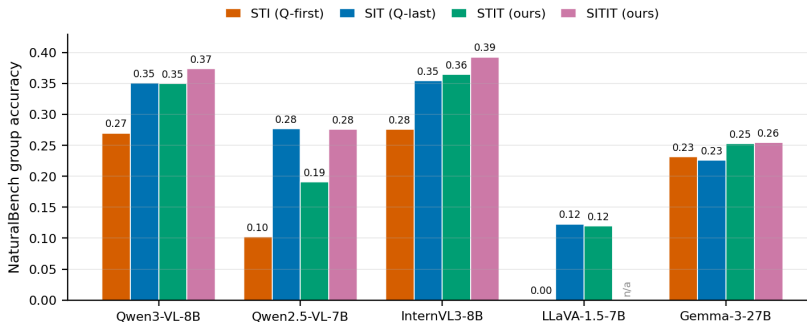


Fig. 6: The position ladder, across models. NaturalBench group accuracy for the five VLMs. On the four models with a paradox, accuracy climbs from question-first (STI) to question-last (SIT) to echoing (STIT) to image re-representation (SITIT); the exceptions are honest and visible (Qwen2.5-VL under-recovers at STIT; Gemma-3 has no STI-SIT gap but still gains from echoing; LLaVA-1.5 is single-image, so SITIT is n/a). Exact per-metric numbers for all models are in Tab. 8.

Table 4: Open-ended VQA (Qwen3-VL-8B, 2,000 VQAv2 validation questions, official soft-accuracy over free-form generation). The position ladder survives unconstrained output: question-first trails question-last (paired bootstrap $\Delta = +0.023$, 95% CI $[+0.011, +0.035]$) and image re-representation is best. Best in **bold**.

	STI	SIT	STIT	SITIT
VQA soft-acc	0.811	0.834	0.821	0.838

system-plus-question on Qwen3-VL), so the variants sit at opposite ends. Question echoing (STIT) adds only a second question, about 27 tokens (+1%), leaving cost essentially unchanged; image re-representation (SITIT) adds a second image and nearly doubles the prefill ($1.98\times$ cache, $3.9\times$ attention FLOPs). Accuracy and cost rank the same way: STIT is the near-free default, and SITIT buys its extra accuracy with double the prefill. Image tokens dominate on every VLM we tested, so this picture is general.

Table 5: Prompt cost per ordering (Qwen3-VL-8B, mean over 30 NaturalBench prompts). A VLM prefills the full prompt, so cost tracks token count: KV-cache scales linearly, self-attention prefill quadratically. Question echoing (STIT) is within 1% of the single-pass orderings; image re-representation (SITIT) roughly doubles the prefill.

Ordering	Image tok	Text tok	Total	KV-cache	Attn. prefill
STI / SIT	2,305	65	2,370	$1.0\times$	$1.0\times$
STIT (ours)	2,305	93	2,398	$1.01\times$	$1.02\times$
SITIT (ours)	4,610	96	4,706	$1.98\times$	$3.94\times$

Table 6: Ablations on NaturalBench (Qwen3-VL-8B, group accuracy). Adding image tokens or pre-answer compute does not close the question-first gap; a post-image question does.

Group	Variant	Group acc
Not more tokens	STI (baseline)	0.270
	STI + img copies $\times 2 / \times 3$	0.281 / 0.280
	STI + mean-resize (fewer)	0.258
Not more compute	STI + space (5/40/120 pad)	0.266 / 0.267 / 0.266
	STIT + chain-of-thought	0.332
Question content matters	STIT (full post-image question)	0.350
	STIT + short cue (same position)	0.292
Ordering ladder	SIT (question-last)	0.351
	STIT (echo question)	0.350
	STITI (echo image, STI units)	0.354
	STITIT (STITI + final Q)	0.352
	SITIT (echo image, SIT units)	0.374
	SITIT-R (SITIT, 2nd image rev.)	0.374

7 Ablations: Isolating the Active Ingredient

The mechanism claims the gap is about *where the question sits*, not about token count or compute. We test the claim by trying to close the gap with everything except a post-image question (Tab. 6; Qwen3-VL-8B, NaturalBench group accuracy).

It is not more image tokens. Duplicating the image under STI lifts group accuracy only marginally (to about 0.28), and shrinking the image to fewer tokens (mean-resize) lowers it to 0.258.

It is not more pre-answer compute. Inserting 5 to 120 padding tokens after the question does not help (about 0.266), and chain-of-thought on top of echoing does not add either (0.332 vs. 0.350).

The question content is what must be adjacent. A full post-image question restores accuracy (STIT 0.350); a short cue in the same position recovers far less (0.292). The post-image copy must restate the question, not merely mark where it ended.

Repeating the image helps only in the training order. The natural way to extend question echoing with a repeated image is STITI: echo the question around the image, then repeat the image. It underperforms SITIT (0.354 vs. 0.374) even though the two contain *identical* content and differ only in how each image and question are paired: SITIT uses the image-first units of SIT, the format the model is trained with, whereas STITI uses the question-first units of STI. The

gap is not adjacency: appending a final question to STITI (STITIT) does not help (0.352). Re-presenting the image pays off only when the repeated units follow the in-distribution image-first order.

Finding. The lever is a restated question adjacent to the answer, not token count or compute: more image tokens (0.281), fewer (0.258), and padding (0.266) leave the gap open, while a full post-image question closes it (0.350; a short cue does not, 0.292). Re-presenting the image helps only in the training format: SITIT beats content-identical STITI by two points (0.374 vs. 0.354).

8 Discussion

Two mechanisms, two positions. Our results separate two effects that prompt design conflates. Putting the question first does create top-down visual steering, but a decoder-only VLM reads its answer mostly from tokens adjacent to the answer position, so a stranded question is barely consulted and the steered features are decoded into an early, image-anchored guess. Question-last wins not because it steers vision better (it steers less) but because the question sits where the answer can use it. Echoing gets both: steer with the pre-image copy, read out with the post-image copy.

Implications beyond prompting. Three follow. *For benchmarking:* reported VQA numbers depend on an undocumented ordering choice worth up to 17.5 points; protocols should fix and report it. *For agentic systems:* pipelines that state the task before attaching visual context run the worst ordering, and a one-line echo repairs them. *For training:* the in-distribution effect (Sec. 7) suggests order-robustness is learned; mixing section orders during instruction-tuning may remove the paradox at the source.

Limitations. We study open VLMs up to 27B parameters on benchmarks dominated by short-answer questions; closed models and long-form generation beyond VQAv2 remain future work. The logit lens is approximate, and activation patching would localize the circuit beyond our knockout. Re-presentation adds the token cost of a second image. Gemma-3-27B, our only model without the paradox, is also our largest and our only one with bidirectional image attention, so we cannot say whether the paradox fades with scale, with bidirectional attention, or both; its bidirectional attention and pretraining may already blunt the read-out failure that echoing targets. Echoing helps it regardless; separating these causes needs models that vary scale and attention independently.

Conclusion. The question-first paradox is a positional read-out failure: the ordering that best steers perception strands the question where the answer cannot read it. The fix is a prompt edit: put the question where it can both steer and be read, and give the decoder a second look at the image.

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A Supplementary: experimental setup and reproducibility

Models. We use the public instruction-tuned checkpoints Qwen3-VL-8B, Qwen2.5-VL-7B, InternVL3-8B, LLaVA-1.5-7B, and Gemma-3-27B, driven by a single order-aware input builder so that orderings differ only in the arrangement of identical tokens. LLaVA-1.5 accepts a single image, so the image-representation orderings (SITIT, SITIT-R) do not apply to it.

Prompt. The fixed system prompt is, verbatim:

A chat between a curious user and an artificial intelligence assistant.
The assistant gives helpful, detailed, and polite answers to the human’s
questions.

Yes/no questions append “Answer the question using only Yes or No.”; two-alternative items append “Answer with only the option letter (A or B).” Decoding is greedy with a 16-token budget, and the prediction is the first Yes/No (or option-letter) token.

Data. NaturalBench (1,900 groups of four (image, question) pairs), POPE (9,000 questions), Winoground (400 groups), and an open-ended split of VQAv2 (2,000 validation questions). Group accuracy counts a NaturalBench or Winoground group correct only if all four of its pairs are.

Open-ended VQA scoring. On VQAv2 the model generates a free-form answer (greedy, no options given). We report the official VQA soft-accuracy: after the standard answer normalization (lowercasing, punctuation, and digit/article handling on both the reply and the ten human answers), a prediction that agrees with k of the ten annotators scores $\min(k/3, 1)$, averaged over the leave-one-out annotator subsets and over questions. Because free-form generation returns a phrase rather than a bare token, we treat a ground-truth answer as produced if it appears in the reply as a whole word (so “no” does not match “snow”); this whole-word containment is a small relaxation of the exact match used with short-answer models.

Significance. For adjacent orderings we test the paired difference in per-group correctness with a two-sided exact McNemar test and a paired bootstrap 95% CI (5,000 resamples).

Mechanistic probes. The perception and read-out probes read a single eager-attention forward pass per (image, question) pair; unless noted they run on 150 disagreement pairs (question-last correct, question-first wrong). Disagreement pairs are chosen deliberately: they are the cases where ordering alone changes the answer, so they act as a microscope on the mechanism, and the per-layer separations are illustrative of it rather than population-average effect sizes. The causal knockout that anchors our conclusion (Sec. 4.3) instead uses 250 outcome-independent yes/no pairs, so the mechanism does not rest on this selection. Code and a project website are included in the supplementary material.

Table 7: Paired significance for the gaps of Tab. 1 and the echoing ladder. Δ is the gap on the benchmark’s correctness unit (NaturalBench and Winoground per-group, all four (image, question) pairs correct; POPE per-question); the 95% CI is a paired bootstrap (5,000 resamples) and p a two-sided exact McNemar test on the discordant pairs. The paradox (STI $\rightarrow$ SIT) is significant everywhere except on Gemma-3.

Comparison	Δ	95% CI	McNemar p
<i>Question-first paradox</i> (STI $\rightarrow$ SIT)			
<i>NaturalBench</i> (group)			
Qwen3-VL-8B	+0.081	[+0.058, +0.103]	$<10^{-4}$
Qwen2.5-VL-7B	+0.176	[+0.155, +0.196]	$<10^{-4}$
InternVL3-8B	+0.079	[+0.054, +0.105]	$<10^{-4}$
LLaVA-1.5-7B	+0.123	[+0.109, +0.138]	$<10^{-4}$
Gemma-3-27B	−0.006	[−0.026, +0.015]	.620
<i>POPE</i> (per-question)			
Qwen3-VL-8B	+0.020	[+0.015, +0.025]	$<10^{-4}$
Qwen2.5-VL-7B	+0.056	[+0.050, +0.063]	$<10^{-4}$
InternVL3-8B	+0.053	[+0.046, +0.060]	$<10^{-4}$
LLaVA-1.5-7B	+0.371	[+0.358, +0.385]	$<10^{-4}$
Gemma-3-27B	+0.005	[−0.001, +0.012]	.124
<i>Winoground</i> (group)			
Qwen3-VL-8B	+0.095	[+0.050, +0.140]	$<10^{-4}$
Qwen2.5-VL-7B	+0.185	[+0.145, +0.225]	$<10^{-4}$
InternVL3-8B	+0.210	[+0.163, +0.258]	$<10^{-4}$
LLaVA-1.5-7B	+0.048	[+0.028, +0.070]	$<10^{-4}$
Gemma-3-27B	−0.005	[−0.052, +0.043]	.916
<i>Echoing ladder</i> (NaturalBench group)			
Qwen3-VL-8B: SIT $\rightarrow$ STIT	−0.001	[−0.022, +0.020]	1.000
Qwen3-VL-8B: STIT $\rightarrow$ SITIT	+0.024	[+0.005, +0.043]	.016
Gemma-3-27B: SIT $\rightarrow$ STIT	+0.026	[+0.006, +0.046]	.014
Gemma-3-27B: STIT $\rightarrow$ SITIT	+0.003	[−0.015, +0.021]	.819

B Supplementary: full per-metric results and controls

Paradox significance. Table 7 reports the paired significance of every gap in Tab. 1, plus the echoing ladder steps on the two primary models. The question-first paradox is significant on every model and benchmark except Gemma-3, the one model where the ordering does not matter.

NaturalBench sub-metrics. Table 8 expands the NaturalBench column of the main results table (Tab. 3) into all four official metrics: Group (all four (image, question) pairs in a group correct), Question (both images correct for a question), Image (both questions correct for an image), and Pair (a single (image, question) pair correct). The position ladder of the main text holds on every sub-metric for every model.

Random-permutation control (SITIT-P). Replacing the second image copy’s reversal with a random permutation of its patches (2D positional indices preserved)

Table 8: NaturalBench (1,900 groups), all four metrics. The ordering ladder of the main text ($STI < SIT \approx STIT < SITIT$) holds on every sub-metric, except on Qwen2.5-VL, where question echoing alone under-recovers and the image copy is needed to match question-last. Best per model in **bold**.

Model	Ordering	Group	Question	Image	Pair
Qwen3-VL-8B	STI (Q-first)	0.270	0.523	0.560	0.753
	SIT (Q-last)	0.351	0.600	0.621	0.792
	STIT (ours)	0.350	0.593	0.622	0.790
	SITIT (ours)	0.374	0.619	0.643	0.804
	SITIT-R (ours)	0.374	0.617	0.643	0.803
Gemma-3-27B	STI (Q-first)	0.232	0.490	0.532	0.732
	SIT (Q-last)	0.226	0.491	0.526	0.733
	STIT (ours)	0.253	0.516	0.551	0.746
	SITIT (ours)	0.255	0.513	0.546	0.746
	SITIT-R (ours)	0.252	0.506	0.540	0.740
Qwen2.5-VL-7B	STI (Q-first)	0.102	0.313	0.336	0.649
	SIT (Q-last)	0.277	0.542	0.562	0.760
	STIT (ours)	0.191	0.451	0.477	0.718
	SITIT (ours)	0.276	0.532	0.551	0.757
InternVL3-8B	STI (Q-first)	0.276	0.530	0.568	0.756
	SIT (Q-last)	0.355	0.605	0.627	0.796
	STIT (ours)	0.365	0.611	0.633	0.798
	SITIT (ours)	0.393	0.631	0.649	0.808
LLaVA-1.5-7B	STI (Q-first)	0.000	0.017	0.035	0.503
	SIT (Q-last)	0.123	0.365	0.413	0.669
	STIT (ours)	0.120	0.357	0.420	0.663

matches both forward and reversed re-representation on all three benchmarks and both models: NaturalBench group 0.380 / 0.249, POPE accuracy 0.888 / 0.834, and Winoground group 0.403 / 0.312 (Qwen3-VL / Gemma-3), each within noise of SITIT. The gain of re-representation is the second whole-image read itself, not any particular order of the second copy.

C Supplementary: the reversal follows the mask (SITIT-R)

Does the second image copy need to be *reversed*? The mechanism says no, and it says so differently for each model, which doubles as a check. Forward repetition already supplies whole-image context, so reversing the second copy’s patch order and positional indices (SITIT-R) should add only scan-order diversity: a small gain for a decoder that is causal over image tokens, nothing for one that is already bidirectional. Reading the attention masks from each model’s code, Qwen3-VL is causal over image tokens and Gemma-3 is bidirectional within each image block (Fig. 5). The prediction holds: reversal helps Qwen3-VL slightly (Winoground

group 0.403 to 0.410) and slightly hurts Gemma-3 (0.318 to 0.298). The probes locate the reversal effect in perception, not read-out: SITIT-R matches SITIT on answer-to-question attention and answer emergence (final $P(\text{correct})$ 0.85 vs. 0.84). A final control replaces the reversal with a *random* permutation of the second copy with 2D positions preserved (SITIT-P): it matches forward and reversed re-presentation on all three benchmarks and both models (Sec. B), so the gain is the second whole-image read itself, not any particular order of the second copy.

Finding. The mechanism predicts the sign of the reversal effect per model, and the prediction holds: reversal helps the causal Qwen3-VL decoder slightly (Winoground 0.403 to 0.410) and not the already-bidirectional Gemma-3 (0.318 to 0.298); the sign tracks the mask read from each model’s code. A random second-copy order (SITIT-P) matches both, so the gain is the second whole-image read, not any particular order of it.

D Supplementary: what the attention knockout does and does not remove

The causal intervention of Sec. 4.3 deserves one clarification, because a natural reading is that forbidding the answer to attend to the question should leave the model no way to answer at all. It does not, for two reasons.

The question stays in the prompt. We do not delete the question or mask it globally. We add a pre-softmax $-\infty$ only to the entries of the attention matrix that connect the *answer position* (the final token, in the reported “last” scope) to the *question-token columns*, at every text layer. Every other position still attends to the question freely; the question is still embedded and still shapes the rest of the computation. The answer-suffix (“... Answer Yes or No:”) is untouched, so the model always emits a well-formed yes/no token. The knockout changes *which* answer the model gives, not *whether* it answers.

The question’s content still reaches the answer, indirectly. In a transformer a token’s meaning does not live only in its own column. Over the early layers, attention copies the question’s content forward into the hidden states of the tokens that sit between the question and the answer (the trailing punctuation and the answer-suffix tokens). Those relay tokens attend to the question freely, since the knockout touches only the answer position’s edges. The answer token can then attend to the relay tokens, so the question’s content reaches it through a two-hop path (question $\rightarrow$ relay token $\rightarrow$ answer) and through the residual stream, even though its direct edge is cut. This forward movement of information is a well-documented property of transformer computation.

What the number therefore measures. The knockout isolates the *marginal* causal contribution of the answer’s *direct* read of the question, on top of whatever leaks through relays; it is not a measure of total question access. This is why

severing an edge the answer does use lowers accuracy only partially (question-last $0.588 \rightarrow 0.532$, toward chance on the question-dependent items) rather than to random output, and it is why the reportable claim is the *dissociation* (the direct edge is causal under question-last and inert under question-first, where the answer instead reads the image), not a full-collapse claim. The same relay effect is why the broader “downstream” knockout, which also severs the relay tokens, behaves less cleanly and is reported only as corroboration.

E Supplementary: the perception probe, mapped

Figure 7 maps the per-patch steering measurement summarized in Sec. 4.1: only a question that precedes the image moves the patch encodings, and echoing reproduces question-first’s steering exactly.

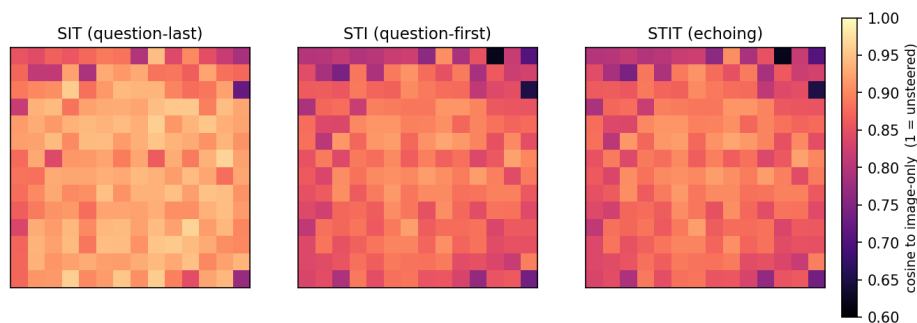


Fig. 7: Steering, measured, across orderings. Per-patch cosine between image-token hidden states (layer 18) and the image shown alone, averaged over Natural-Bench images (darker = moved further). Question-last (SIT) barely steers the image (mean 0.91); question-first (STI) steers it markedly (mean 0.86, down to 0.62); echoing (STIT) is identical to STI (cosine 1.000) because the causal mask hides the post-image question from the patches. Steering comes only from a question that precedes the image, so echoing acts purely at read-out.

F Supplementary: the gap scales with image-token count

Figure 8 gives the full resolution sweep behind Sec. 4.4. We re-render Natural-Bench at a range of resolutions, which sets the vision-token count, using only the 916 images whose short side exceeds the largest rendered resolution (784px) so that every point is a genuine downscale and no image is upsampled. Question-last (SIT) saturates near 0.35 group accuracy once the image is legible; question-first (STI) stays lower and keeps climbing as better perception partly offsets its read-out penalty. The SIT–STI gap widens with token count in the distance-dominated regime (0.05 at 64 tokens to 0.08 at 324), then plateaus as SIT saturates, matching the read-out account’s prediction in direction and shape.

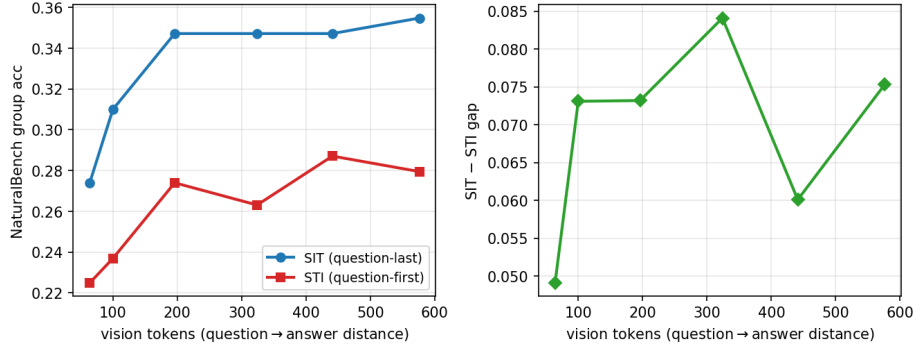


Fig. 8: The question-first gap scales with the number of image tokens. NaturalBench group accuracy (Qwen3-VL, 916 images with short side ≥ 784 px). *Left:* question-last (SIT) saturates near 0.35 once the image is legible; question-first (STI) stays lower and keeps climbing. *Right:* the SIT–STI gap widens with token count (0.05 at 64 tokens to 0.08 at 324), then plateaus as SIT saturates. More image tokens push the question further from the answer under STI, while SIT keeps it adjacent.

G Supplementary: is the question-first rewrite localized?

The main text shows that question-first prompting steers the image representation (Sec. 4): intermediate patch encodings become more question-relevant. A natural question is whether this rewrite is *spatially selective*, that is, whether asking about one object rewrites the patches on that object more than others. We test this directly on a single image with two questions, and find that the rewrite is real but *diffuse*: it is a global, question-conditioned shift of the whole visual field rather than a spotlight on the queried object. We show this on a single image, as an illustration of *how* the steering distributes rather than a claim about how typical the pattern is. The diffuseness may also be architectural: Qwen3-VL injects visual features into the decoder at multiple depths (its DeepStack design [21]), which spreads image information across layers and could blur an object-localized rewrite into a global one. We read the result as suggestive and do not build on it.

Setup. We use one image (a living room with a television showing football and a sleeping cat) and two questions, q_1 = “What sport is on TV?” and q_2 = “What is the cat doing?”. For an ordering we take the image-token hidden states $h(q)$ at every layer and, per patch, measure $\cos(h(q_1), h(q_2))$: how much does *which question you ask* change each visual patch. Under an image-first ordering the decoder mask prevents the image tokens from ever attending to the (later) question, so the image representation cannot depend on the question; under question-first (STI) the question precedes the image and can rewrite it.

The representation is question-dependent only under question-first. Figure 9 confirms the mask prediction exactly. Image-first keeps $\cos(h(q_1), h(q_2)) = 1.000$ at

every layer: the visual representation is completely question-invariant. Question-first (STI, and equivalently question-echoing STIT) drives the cosine down with depth to 0.825 at the final layer: the same image is encoded differently depending on what is asked. This is the representation-level counterpart of the steering result in Sec. 4.

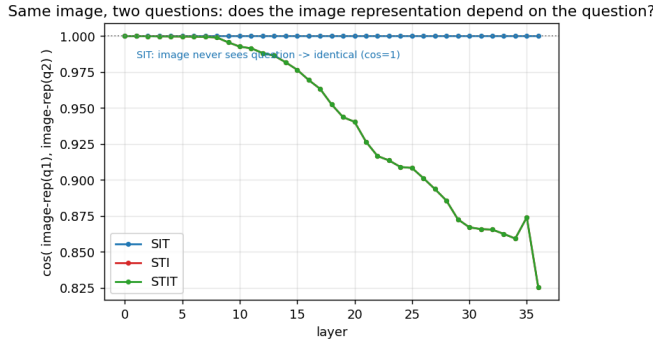


Fig. 9: Same image, two questions. Per-layer cos between the image-token representations under q_1 and q_2 . Image-first (blue) stays at 1.000 (the image never sees the question, so its encoding is question-invariant); question-first STI/STIT (red under green, identical curves) falls to 0.825 (the question rewrites the image). This is a scalar, not a spatial, effect (see Fig. 10).

But the rewrite does not localize on the queried object. We asked whether the question-first rewrite concentrates on the relevant object (the TV for q_1 , the cat for q_2). It does not. Writing $e_q = h_{\text{STI}}(q) - b$ for the change each question makes relative to the image-only baseline b , two facts emerge. First, the two questions’ effect vectors are 92–96% the same vector: the shared, question-agnostic “a question is present” component dominates, and only about 5% of e_q is specific to which question was asked. Consequently the per-question perturbation maps $\|e_{q_1}\|$ and $\|e_{q_2}\|$ are nearly identical (spatial correlation 0.98). Second, the raw per-patch magnitude is dominated by a few outlier “massive-activation” hidden dimensions (per-patch coefficient of variation 6.8, salt-and-pepper speckle); standardizing each dimension removes this (coefficient of variation 0.69). Even after removing both the shared component and the outlier dimensions, the residual question-specific perturbation is spatially diffuse (Fig. 10), not concentrated on the queried object; a gradient-based (Grad-CAM) attribution of the answer token is likewise dominated by background and border patches. The question-first steering is a broad rewrite of the visual field, consistent with the main-text account in which the paradox lies in downstream read-out rather than in mislocalized perception. The scalar curve of Fig. 9, not a per-patch heatmap, is the faithful visualization of this effect.

A semantic logit-lens readout does localize. The diffuseness above is a property of the raw representation geometry, not of the model’s percept. Projecting each

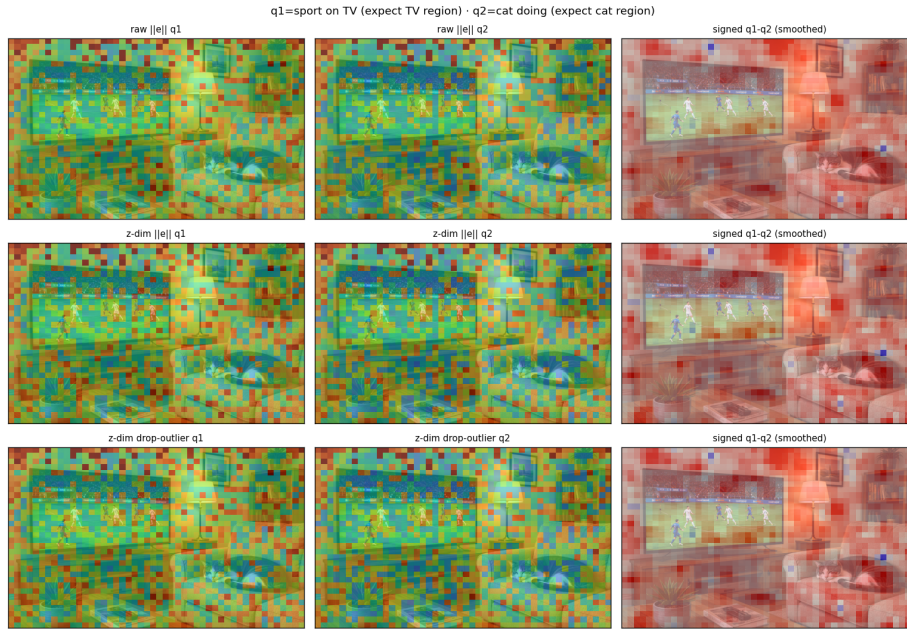


Fig. 10: Per-patch question-first perturbation of the image representation (layer 18). Rows: raw magnitude $\|e_q\|$ (top, dominated by outlier-dimension speckle), per-dimension standardized (middle), and standardized with the 20 largest-variance dimensions dropped (bottom). Columns: q_1 magnitude, q_2 magnitude, and the signed difference $\|e_{q_1}\| - \|e_{q_2}\|$ (smoothed; red = q_1 rewrote more, blue = q_2). After denoising, the q_1 and q_2 maps remain almost identical (correlation 0.98) and the signed difference is a near-uniform field rather than a spotlight on the TV (q_1) or the cat (q_2): the rewrite is diffuse, not object-localized.

image patch through the logit lens (final norm and unembedding) and summing the probability on a concept’s token set recovers a localized, object-level readout. Figure 11 plots the per-patch probability of cat-words minus sport/TV-words at layer 28: the television reads as sport/TV (blue) and the sleeping cat reads as cat (red), with each concept’s mass concentrated inside its object box (cat $8.6\times$, sport/TV $5.4\times$ more mass inside the box than outside). Decoding each patch to its top vocabulary token tells the same story in words (Fig. 12): the crowd patches read *Barcelona/soccer/stadium*, the players *jerseys*, the cat *asleep/striped*, the potted plant *ceramic/plant*. The readout is also question-sensitive under question-first: the concept mass on each object ROI changes between the two questions under STI, whereas under image-first it is identical (Fig. 13). The question-first rewrite is thus diffuse in the residual stream yet semantically localized once projected to token space. This is exactly what the read-out account of the main text predicts: perception localizes objects correctly, so the paradox is a downstream failure to read the correctly perceived answer, not a failure to perceive.

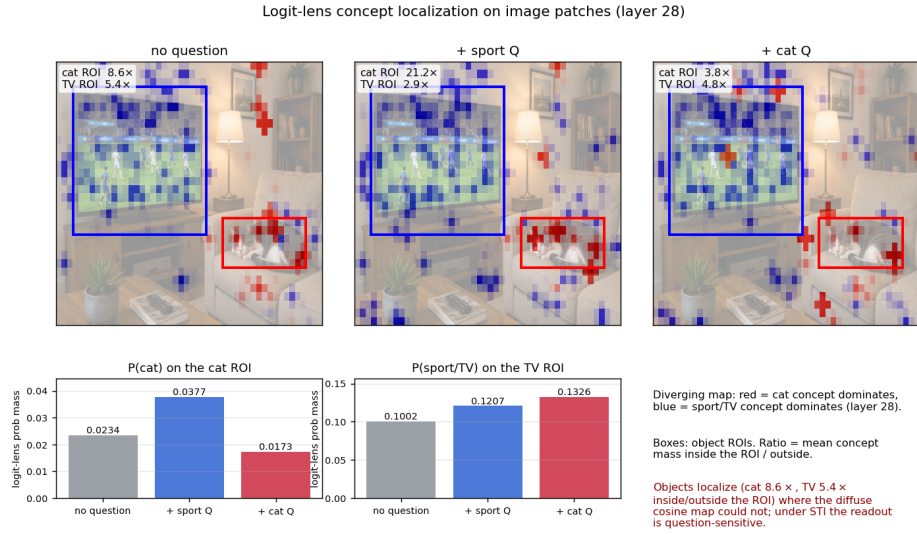


Fig. 11: Per-patch logit-lens concept localization (layer 28). Top: the diverging map $P(\text{cat}) - P(\text{sport/TV})$ overlaid on the image (red: cat concept dominates; blue: sport/TV concept dominates), for the image-only baseline and question-first under each question; boxes are the object ROIs and the corner text is the inside/outside mass ratio. Bottom: raw logit-lens probability mass inside each object’s ROI across conditions. Unlike the diffuse cosine map (Fig. 10), the semantic readout localizes each object, and under question-first the readout on each object is question-sensitive.

The read-out is question-dependent only under question-first. Figure 13 makes the ordering contrast explicit on the same image. Under image-first (SIT) the image tokens precede the question and are masked from it, so the per-patch logit-lens read-out is bit-identical for the two questions ($\Delta P = 0$ exactly). Under question-first (STI) the question precedes the image and moves the read-out: the concept mass on each object ROI differs between the two questions ($|\Delta P| > 0$). We report the magnitude of this question-effect rather than its sign, which is not stable on a single image; the robust, mask-level fact is that question-first makes the visual read-out question-dependent while image-first freezes it. This is the localized, semantic counterpart of the scalar cosine result (Fig. 9).

H Supplementary: system prompt placement (IST vs SIT)

Every ordering in the main text prepends the fixed system message. Here we justify that default by moving only the system block: IST (Image, System, Task) leads with the image, while SIT (System, Image, Task) leads with the system prompt. The two are anagrams that keep the question in the same question-last position, so any difference isolates the effect of *where the system prompt sits* relative to the image.



Fig. 12: Logit-lens patch decodings (Qwen3-VL-8B, layer 28). Each callout gives the top vocabulary tokens that a patch decodes to under the logit lens. The patches decode to their objects: the television crowd to *Barcelona/soccer/stadium*, the players to *jerseys*, the lamp to *cozy/lighting*, the cat to *asleep/striped*, and the potted plant to *ceramic/plant*. The visual representation carries localized, human-readable object identity, which is what the read-out stage must access; the question-first paradox is a failure to read this out, not to form it.

System-first helps. On Qwen3-VL-8B, placing the system prompt before the image (SIT) beats image-first (IST) on all three benchmarks (Tab. 9): NaturalBench group accuracy rises $+0.011$ ($0.339 \rightarrow 0.350$), POPE accuracy $+0.002$, and Winoground group accuracy $+0.038$ ($0.280 \rightarrow 0.318$). The gains are small but uniformly positive across datasets and metrics, which is why we prepend the system message in every ordering we report; it also matches the standard chat-template convention the model is trained with.

Relation to the steering analysis. This connects to Fig. 7. Under IST the image is the very first content, so under the causal mask its patch encodings are identical to the image-only baseline (cosine exactly 1.000). Under SIT the system prompt precedes the image, so the image tokens attend back to it and their encoding moves slightly off image-only (mean cosine 0.91). The accuracy numbers show that this system-conditioned shift is benign to mildly helpful on Qwen: perturbing the image with a task-agnostic system prefix does not cost accuracy, unlike the question-first rewrite, whose damage is a read-out failure rather than a perceptual one (Sec. 4).

Finding. Prepending the system prompt (SIT) rather than leading with the image (IST) gives a small, consistent accuracy gain on Qwen3-VL-8B across all three benchmarks (up to $+0.038$ Winoground group accuracy), so we default to system-first. The system prefix perturbs the image encoding only mildly (cosine 0.91 vs. 1.000) and benignly, unlike the question-first rewrite.

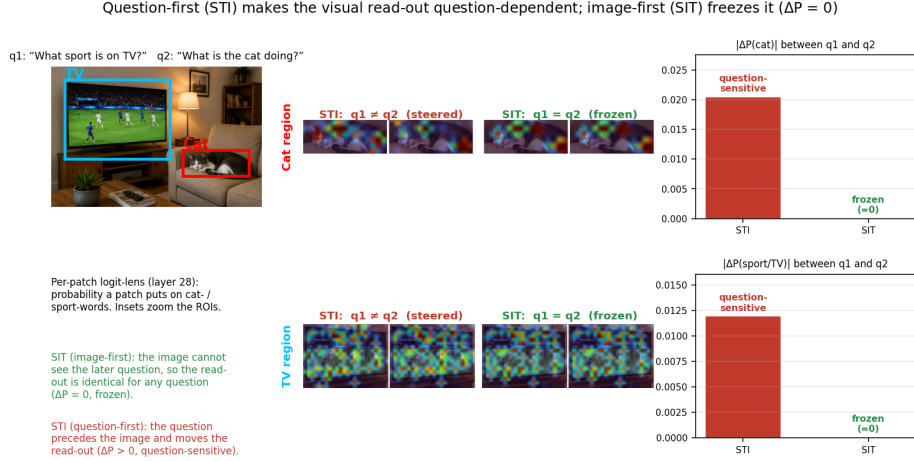


Fig. 13: Question-first steers the visual read-out; image-first cannot. Same image, two questions (q_1 sport, q_2 cat). Insets zoom the per-patch logit-lens concept probability on the cat and TV ROIs; bars give the question-effect magnitude $|\Delta P|$ between the two questions on each ROI. Under STI the read-out differs between questions ($|\Delta P| > 0$, question-sensitive); under SIT the image cannot attend to the later question, so the read-out is identical ($\Delta P = 0$, frozen). We show the magnitude of the question-effect, not its sign, which is not stable on one image. Single image, layer 28; illustrative of the mechanism, not a dataset-level measurement.

Table 9: System prompt placement on Qwen3-VL-8B, question held question-last. SIT (system first) vs IST (image first); $\Delta = \text{SIT} - \text{IST}$. System-first helps on every benchmark. NaturalBench and Winoground are group accuracy, POPE is accuracy.

Benchmark (metric)	IST	SIT	$\Delta = \text{SIT} - \text{IST}$
NaturalBench (Group)	0.339	0.350	+0.011
POPE (Acc)	0.888	0.891	+0.002
Winoground (Group)	0.280	0.318	+0.038